DSE4101
Capstone Project in Data Science and Economics I

Course Introduction

Semester 1 AY 2026/2027



College of Humanities
and Sciences

Course Coordinators

Ta-Cheng HUANG

- Email: tchuang@nus.edu.sg
  - Use email for **administrative and logistical** questions, *not substantive* course questions.
  - Any e-mail to me should be initiated from your NUS e-mail with the course code, i.e., **DSE4101**, in the subject line.
- Office: AS2-0507
- Office Hours: By appointment

What Is a Capstone Project?

- A capstone project is a culminating academic experience where students apply their knowledge to real-world problems or challenges.
- It encourages integration of skills and knowledge acquired throughout the program.
- Provides an opportunity to showcase creativity, critical thinking, and problem-solving abilities.
- Students often select a project based on their individual learning profiles and interests.
- Typically involves independent identification of topics or problems to work on.

Our Directed Capstone Projects

- Students will select one of three domains:
  - Digital Currency,
  - FinTech, or
  - Digital Economy.
- Each domain offers two predefined topics provided by the course coordinators. **The list of topics, along with brief introductions and references, is available on Canvas.**
  - Students may propose a new topic, subject to instructor approval.
- This structured approach aligns with course objectives and enables coordinators to offer targeted guidance and support.
- It fosters focused exploration and promotes a deeper understanding within the chosen domain.

Course Structure

How This Course Works

Weeks 1 – 6

1. **Course Briefing** *11 Aug*
2. **Indicate Domain Preferences** *14 Aug*
   via `https://forms.cloud.microsoft/r/RmL8JxCQwb`
   Students will be informed of their group assignment on *15 Aug*.
   Each group will then select a topic by *18 Aug*.
3. **Phase 1 Proposal** *25 Aug*
   A short written proposal covering the project's motivation, list key literature, potential data sources, outline each member's responsibilities, and provide a basic work plan.
4. **Mid-Term Presentation** *15 Sept*
   Each group studies its chosen topic and presents its understanding of the relevant literature and the conceptual framework to the class.
5. **Discussant Duties**
   - Assigned to review a peer presentation.
   - Submit a **Mid-Term Review Report** summarizing findings and feedback by *22 Sept*

How This Course Works

Weeks 7 – 13

1. **Phase 2 Proposal** *29 Sept*
   Each group begins its journey to apply what they've learned to a real-world dataset and submits an updated proposal covering
   - Chosen dataset & justification
   - Research questions & methods
   - Expected outcomes, task allocation, timeline
2. **Final Presentation** *3 Nov*
   Present research findings, showing application of theory to real-world data.
3. **Final Report** *6 Nov*
   Detailed write-up including motivation, literature, data, methodology, findings, and conclusions.
4. **Discussant Duties** *3 Nov & 13 Nov*
   - Assigned to review a peer presentation. Ask thoughtful questions, give constructive feedback. *3 Nov*
   - Submit a **Final Review Report** Provide summary and critique of research design, data usage, methodology, and conclusions. *13 Nov*

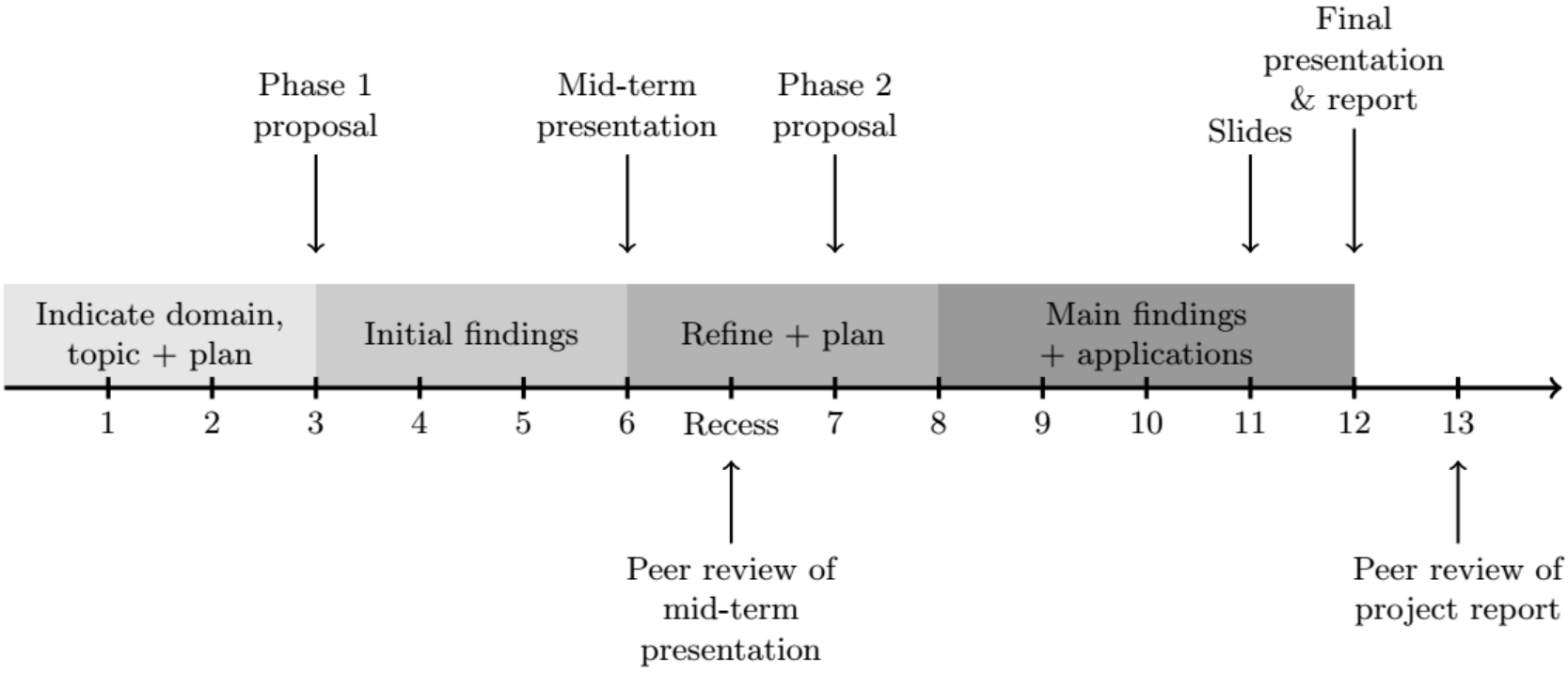
How This Course Works (Chronological Order)

The table below summarizes the key dates.

Week	Date	Time	In-Class Event[a]	Task[b]
1	11 Aug	6–7 pm	Course briefing	
1	14 Aug			Indicate domain preferences
2	18 Aug			Topic selection
3	25 Aug			Phase 1 proposal due
6	13 Sept			Mid-term presentation slides
6	15 Sept	6–10 pm	Mid-term presentation	
R	22 Sept			Review report on a presentation
7	29 Sept			Phase 2 proposal
12	1 Nov			Final presentation slides
12	3 Nov	6–10 pm	Final project presentation	
12	6 Nov			Final group project report
13	13 Nov			Review report on a group project

[a] Meeting in person at AS2-0312.

[b] Tasks are due by 11:59 pm on the deadline, unless stated otherwise.


Phase 1 proposal
Mid-term presentation
Phase 2 proposal
Slides
Final presentation & report
Indicate domain, topic + plan
Initial findings
Refine + plan
Main findings + applications
1
2
3
4
5
6
Recess
7
8
9
10
11
12
13
Peer review of mid-term presentation
Peer review of project report

Course Policies

Grading Components

Group Components (60%):

- **Proposals (10%):** Two (2) group project proposals together account for 10% of the course grade, with 5% allocated to each.
- **Presentations (25%):** Mid-term and final presentations contribute 10% and 15% to the final grade, respectively. Presentations should articulate the group's learning and findings with an engaging and insightful delivery.
- **Final Report (25%):** Provide a detailed summary of the group's findings and applications of theories and concepts studied.
- **Group-component grades can be adjusted if there is documented evidence that a member has not contributed adequately.**

Grading Components Cont'd

Individual Components (40%):

- **Discussant Duties (10%):** As discussants, students are expected to engage actively during both the mid-term and final presentation sessions by asking thoughtful questions, offering constructive feedback, and responding professionally during peer discussions.
- **Review Reports (30%):**
  - **On a Mid-term Presentation (10%):** The report should evaluate and critique another group's presentation for clarity and insights.
  - **On a Final Report (20%):** The report should summarize the key ideas and findings of the project and offer a thoughtful critique of its research design, data usage, methodology, and conclusions. Students are expected to assess both the technical execution and the clarity of communication.

Grading Components Summary

Components	Weightage
*Group (60%)**	
Phase 1 Proposal	5%
Phase 2 Proposal	5%
Mid-term Presentation	10%
Final Presentation	15%
Group Report	25%
Individual (40%)	
Participation: Discussants	10%
Review Report: Peer Mid-term Presentation	10%
Review Report: Peer Group Project	20%
Total	100%

*Group-component grades can be adjusted if there is documented evidence that a member has not contributed adequately.

Grades

Slides 9 and 10 set out the **only** method that will be used to determine course grades.

If you feel that personal circumstances are affecting your academic performance, or are concerned that your work is not going to earn you the grade that you require for some purpose (e.g., getting/keeping an assistantship or a scholarship, graduating on time), please explore your options and take an appropriate action in good time (see Important Dates below).

Canvas

All course materials and announcements will be posted on Canvas. You should check Canvas on a frequent and consistent basis. However, course materials may be changed by in-class announcements. Anything that is missed because of the absences without university-approved and documented excuses is the student's responsibility.

E-Mail

- It is everyone's responsibility to check their official NUS email frequently and consistently.
- Use email for **administrative and logistical** questions, *not substantive* course questions.
- Write from your NUS email account and put the course code, i.e., **DSE4101**, in the subject line.

University Policies

Important Dates

When a student withdraws from a module between the first day of Week 3 of the instructional period and the last day of the Recess Week, a 'W' grade will be recorded in the transcript. Withdrawals after this period (i.e., from the first day of Week 7 of the instructional period) will result in an 'F' grade, which will be included in the computation of the CAP and will not be eligible for the S/U option.

In summary,

- 'W' Grade takes effect from 24 August to 27 September.
- 'F' Grade takes effect from 28 September.

NUS Academic Calendar:
https://nus.edu.sg/registrar/docs/default-source/calendar/ay2026-2027.pdf

NUS Modular System: https://www.nus.edu.sg/registrar/academic-information-policies/undergraduate-students/modular-system

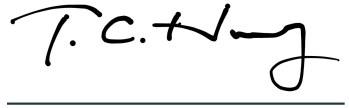
Academic Integrity and AI

- You may use AI tools (e.g., ChatGPT, Claude, Gemini, etc.), but you remain responsible for all submitted work.
- Acknowledge AI use whenever required. Submitting AI-generated work as your own is plagiarism, just like submitting someone else's work as your own.
- Verify and refine AI outputs. AI can make mistakes, and you are responsible for checking accuracy.
- The University has zero tolerance for academic misconduct, including plagiarism, fabrication, copyright infringement, and misuse of AI.

Q&As

Summary

- The syllabus serves as a formal agreement between myself and my students, outlining the course and university policies that we are both expected to uphold with integrity and seriousness.
- It ensures that grades are determined solely by performance in the defined grading components, free from personal considerations, and that all parties respect the boundaries and standards set forth in this document.

Student: _____. Instructor: 

Welcome Aboard

Let the Journey Begin